AN IDEAL CONTAINED IN A PRINCIPAL IDEAL IS RECOVERED AS THE PRODUCT WITH ITS IDEAL QUOTIENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The ideal quotient I : J and the special case I : a. Definition of the ideal quotient, used at proof line 5339. Replaces the quotient by the principal ideal (a) with the more concrete quotient by the element a.
- Divisibility inside the principal ideal (a). Used at proof lines 5340-5341. Writes an element of I contained in (a) as a multiple of a with multiplier in the ideal quotient.

Lemma 0.1. *Let R be a ring. For a principal ideal $J \subset R$, and for any ideal $I \subset J$ we have $I = J(I : J)$.*

Proof. Say $J = (a)$. Then $(I : J) = (I : a)$. Since $I \subset J$ we see that any $y \in I$ is of the form $y = xa$ for some $x \in (I : a)$. Hence $I \subset J(I : J)$. Conversely, if $x \in (I : a)$, then $xJ = (xa) \subset I$, which proves the other inclusion. □

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